

CRM Sales Pipeline Analysis

A Business Analyst Case Study

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Dataset: Kaggle — CRM Sales Opportunities

Tools: Excel · MySQL/MariaDB · Power BI · DAX

Table of Contents

- 1. Executive Summary
- 2. Business Context & Objectives
- 3. Methodology
- 4. Findings
 - 4.1 Agent & Regional Performance
 - 4.2 Pipeline Drop-off & the Quarter-End Pattern
 - 4.3 Product & Sector Revenue
 - 4.4 Account Size vs. Outcome
- 5. Cross-Cutting Synthesis
- 6. Limitations & Future Enhancements
- 7. Appendix: Dashboard Overview

1. Executive Summary

This case study analyzes one year (2017) of CRM sales pipeline data covering 8,800 opportunities, 85 accounts, 7 products, and 35 sales agents across 3 regional offices. The engagement was structured around five client questions covering agent and regional performance, pipeline drop-off, product and sector revenue, account-level outcomes, and any hidden cost drivers.

Headline conclusion: no evidence of business decline was found. Win rates are consistently healthy (60–65%) across every region, product, and sector. The opportunities identified are efficiency and scaling gaps within an otherwise stable business — not signs of an underlying problem.

Two findings stand out as the most significant, and both represent revenue upside rather than risk:

- Quarter-end win-rate spike: every sales agent shows a sharp, consistent jump in win rate during the final month of each quarter (78–83%) against a ~50% baseline the rest of the year — a company-wide pattern with real forecasting implications.
- Account under-penetration: deal size fails to scale with account size. The top 10 accounts hold 5× the revenue of standard accounts but buy only 6.6% larger deals — a clear account-expansion opportunity.

The remaining sections detail the methodology, findings for each client question, and prioritized recommendations.

2. Business Context & Objectives

The analysis was framed around a client brief from Sales Operations leadership, who lacked data-backed answers to recurring questions raised in sales leadership meetings. Five questions defined the scope:

- Which sales agents and regional offices are winning or underperforming, and why?
- Where in the pipeline are deals stalling or being lost?
- Which products and account sectors drive the most (and least) revenue?
- Does account size (revenue, employees) relate to deal outcomes?
- Is anything costing the business money that isn't already known?

The deliverable requested was an interactive Power BI dashboard supported by a written summary that pairs each diagnosis with a concrete recommendation — not observations alone.

3. Methodology

3.1 Data Preparation

The source data consisted of five CSV files (accounts, products, sales_teams, sales_pipeline, and a data dictionary) with no built-in relational keys beyond a single opportunity identifier. Before analysis could begin, the following data quality steps were required:

- Added a surrogate account_id to the accounts table, needed specifically because the table contains a self-referencing subsidiary_of column that benefits from a stable key rather than a text name.
- Identified and corrected a product naming mismatch ("GTXPro" vs. "GTX Pro") between the pipeline and product tables that would otherwise have silently broken revenue joins.
- Diagnosed and resolved a foreign key import failure caused by blank subsidiary_of values being loaded as empty strings rather than NULL — a common but easy-to-miss CSV import issue.
- Verified no duplicate sales agent names existed before treating agent name as a safe join key.

3.2 Analytical Approach

Each finding in this report was validated twice: first with a hand-written SQL query against the cleaned MySQL/MariaDB database, then independently reproduced using DAX measures inside Power BI. Where the two disagreed during development, the discrepancy was treated as a bug to resolve (see the win-rate correction in Section 4.1) rather than a modeling choice to accept quietly.

Two methodological principles were applied consistently:

- Win rate is always calculated as $\text{Won} \div (\text{Won} + \text{Lost})$, excluding still-open Prospecting and Engaging deals from the denominator — including open deals would understate every agent's and region's true conversion rate.
- Where account segmentation required grouping into size tiers, percentile-based splitting (SQL's NTILE window function) was used instead of fixed dollar-range buckets, since account revenue is heavily right-skewed (a small number of very large accounts would otherwise distort equal-width buckets).

4. Findings

4.1 Agent & Regional Performance

Win rate proved to be a poor differentiator at both the regional and product level, sitting in a tight 60–65% band regardless of how the data was sliced. Revenue differences are driven almost entirely by average deal size, not by how often a region or agent wins.

Region	Total Deals	Win Rate	Avg Deal Size	Total Revenue
Central	3,512	62.6%	\$2,054	\$3.35M
West	2,997	63.9%	\$2,482	\$3.57M
East	2,291	63.0%	\$2,639	\$3.09M

East closes the largest deals on average but has the fewest total deals, suggesting a pipeline-volume constraint rather than a skill gap. Central runs the highest volume but the smallest average deal, worth reviewing for under-pricing or over-targeting smaller accounts.

At the agent level, two distinct high-performer profiles emerged rather than one:

- Darcel Schlecht — highest individual revenue (\$1.15M), driven by the largest average deal size (\$3,304) and a faster-than-average sales cycle (45.8 days), not an unusually high win rate.
- Versie Hillebrand — highest win rate on the team (66.7%), but the lowest average deal size (\$1,066), resulting in comparatively modest total revenue.

Across all 35 agents, win rate ranged more widely (55–70%) than region or product ever did — the widest natural variation found anywhere in the dataset at that stage of analysis. Corliss Cosme and Reed Clapper were selected as the team's benchmark: rather than leading on any single metric, both are consistently above-average across win rate, deal size, and revenue, making their approach realistically repeatable across the wider team.

Recommendation: don't manage Darcel and Versie the same way. Study and replicate Darcel's large-account approach; evaluate whether Versie's high closing skill could be redirected toward larger accounts. Use Corliss Cosme and Reed Clapper, not the single top performer, as the coaching benchmark.

4.2 Pipeline Drop-off & the Quarter-End Pattern

Of 8,800 total opportunities, 4,238 (48%) were Won, 2,473 (28%) were Lost, and the remainder were still open (Engaging 1,589; Prospecting 500) at the time of export.

Loss patterns by product and by agent were checked for concentration. Neither showed a meaningful red flag: product-level win rate stayed within the same 60–65% band as regions, and estimated lost revenue by product was found to be driven almost entirely by list price rather than any product losing disproportionately often.

The most significant finding of the project emerged from checking losses and wins against time. Every single sales agent — all 30 with sufficient data — showed a substantial win-rate increase during the final month of each quarter:

Period	Win Rate	Avg. Cycle Length (Won deals)
Quarter-end months (Mar / Jun / Sep / Dec)	78–83%	65.7 days
All other months	~49–57%	38.4 days

The size of the effect varied by agent (from a +9-point gap up to +42 points), but the direction was universal — this is a structural, company-wide pattern, not the behavior of a few individuals. Quarter-end wins also carry meaningfully longer sales cycles, which points toward deals accumulating throughout the quarter and being resolved in a batch near the deadline, rather than being rushed through quickly.

Recommendation: forecast using month-specific win rates, not the single blended 63% average, which represents no real month. Confirm with sales leadership whether deals are deliberately held open and batch-resolved near quarter-end (e.g., for quota or contract/budget cycles) — if so, consider a more even monthly review cadence to reduce end-of-quarter operational strain and improve real-time pipeline visibility.

4.3 Product & Sector Revenue

Both products and account sectors were checked for the same win-rate and deal-size patterns already established at the regional level, and both returned essentially the same conclusion: flat, healthy performance with no standout risk.

Dimension	Win Rate Range	Notable Pattern
Product (7 products)	60.3% – 65.4%	MG Advanced slightly below the norm; not dramatic enough to flag as a concern
Sector (10 sectors)	61.2% – 64.9%	Revenue differences reflect account count per sector, not sector quality

One data quality note: the sector value "technolgy" is misspelled in the source data. It does not create a duplicate-category problem here (no correctly spelled variant exists), but would be worth correcting at the source if this dataset is ever extended.

Given how flat this dimension is, a dedicated dashboard page was deliberately not built for sector — the finding is fully captured here in writing, and a dashboard page built purely to show "no pattern" would add clutter without adding decision-making value.

4.4 Account Size vs. Outcome

This question produced the single strongest, most repeatable finding in the analysis. Accounts were segmented two ways to confirm the pattern held at different scales:

- Whale vs. Standard: the 10 highest-revenue accounts (11.8% of all accounts) hold 40% of total account revenue, averaging 5× the revenue of the remaining 75 accounts.
- Large vs. Small (excluding whales): even among the remaining, less extreme accounts, the top revenue tier averages 9.6× the revenue of the bottom tier.

Segment	Accounts	Avg. Account Revenue	Avg. Deal Size	Win Rate
Whale (Top 10)	10	~\$6,788M	\$2,491	61.2%
Standard	75	~\$1,356M	\$2,335	63.6%

The core finding: in both comparisons, a 5–10× difference in account revenue translated into only a 7–16% difference in deal size. Win rate showed no meaningful difference between segments either. This indicates the sales team is under-penetrating its highest-value accounts — selling them essentially the same size deal as much smaller companies, despite vastly greater purchasing capacity.

Recommendation (two-track, prioritized by return on effort): dedicate high-touch account-expansion resources (named account management, tailored proposals, executive relationship-building) to the 10 whale accounts, where the absolute revenue upside per account is large. For the remaining 75 standard accounts, apply scalable, lower-touch tactics (automated upsell triggers, account-size-aware pricing guidance, broader sales training) rather than 1:1 attention, since per-account payoff doesn't justify the same intensity of effort.

5. Cross-Cutting Synthesis

Viewed together, the four findings form a consistent picture. Win rate is remarkably stable across every dimension tested — region, product, sector, and (mostly) agent — which rules out any single weak link dragging down performance. The two dimensions that did show real, actionable variation were individual agent skill (a genuine but moderate 55–70% spread) and account-level deal sizing (a significant scaling gap).

This is a healthy business with identifiable room to grow, not a business with something broken. The recommended focus areas are where to concentrate improvement effort, not where to intervene in a failing process:

- Forecast and plan around real monthly win-rate patterns, not a single blended average.
- Prioritize account expansion effort on the top 10 accounts, where the dollar upside per account is largest.
- Use Corliss Cosme and Reed Clapper's balanced approach, not a single outlier performer, as the team coaching benchmark.

This conclusion is scoped deliberately: it reflects one year of data from a single CRM export, not a full operational audit. It is offered as a well-evidenced starting point for further conversation with sales leadership, not a final verdict on the business as a whole.

The dashboard's Overview page makes this synthesis visible at a glance: revenue is generated in every one of the ten active months, with the quarter-end pattern documented in Section 4.2 explaining the shape of that trend rather than any underlying instability.

6. Limitations & Future Enhancements

The following items were deliberately scoped out of this phase of the analysis, to prioritize delivering a complete, validated core dashboard over a partially-built set of extras:

- An "Agent Deep Dive" dashboard page combining a regional slicer with the agent scatter plot, for drill-down analysis beyond the current static views.

- A revenue forecast for currently open (Prospecting/Engaging) deals, projected using historical win-rate and deal-size patterns.
- Because this is a synthetic Kaggle dataset, some patterns found here — most notably the quarter-end seasonality — may partly reflect how the dataset was generated rather than genuine business behavior. The recommendation to confirm this with the real sales team applies regardless of the underlying cause, since the forecasting risk (relying on a blended win rate) is the same either way.

7. Appendix: Dashboard Overview

The accompanying Power BI file contains four interactive dashboard pages, each built on a shared data model (accounts, products, sales_teams, sales_pipeline) connected through validated relationships:

- Overview — a landing page with top-level KPIs, a monthly revenue trend confirming no decline over the year, and condensed views of the regional and account-segment findings, alongside a short executive summary.
- Regional Performance — KPI summary, deal-size and revenue breakdowns by region, and the regional benchmark finding.
- Agent Performance — a full 35-agent table and scatter plot (win rate vs. average deal size, sized by revenue), with an agent slicer for individual drill-down.
- Account Segment — whale-vs-standard comparison charts and an account-level scatter plot (revenue vs. average deal size) visualizing the under-penetration finding directly.

All four pages follow a consistent visual identity (the "Veylo" theme: sage green, navy, and gold on a cream background) and a consistent structure of KPI cards, supporting charts, and short diagnosis-plus-recommendation callouts.